Chapter 10: User Interface Design

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Course: Web Programming
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Content

- Bootstrap: Component-Based UI
- Bootstrap components & forms
- TailwindCSS: Utility-First Styling
- TailwindCSS syntax & responsiveness

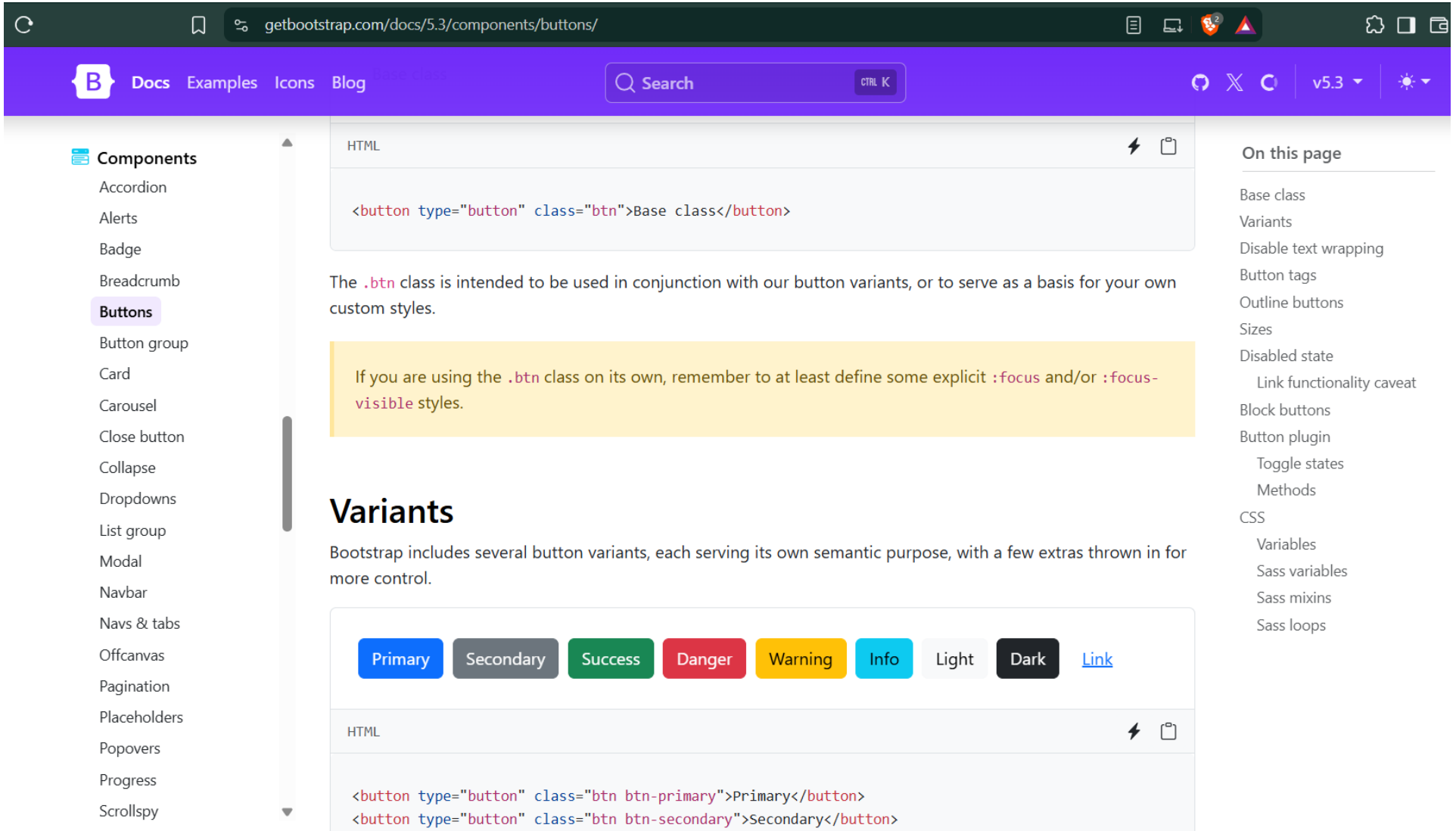
Bootstrap: Component-Based UI

Rapid prototyping with pre-built responsive components

The component-based approach

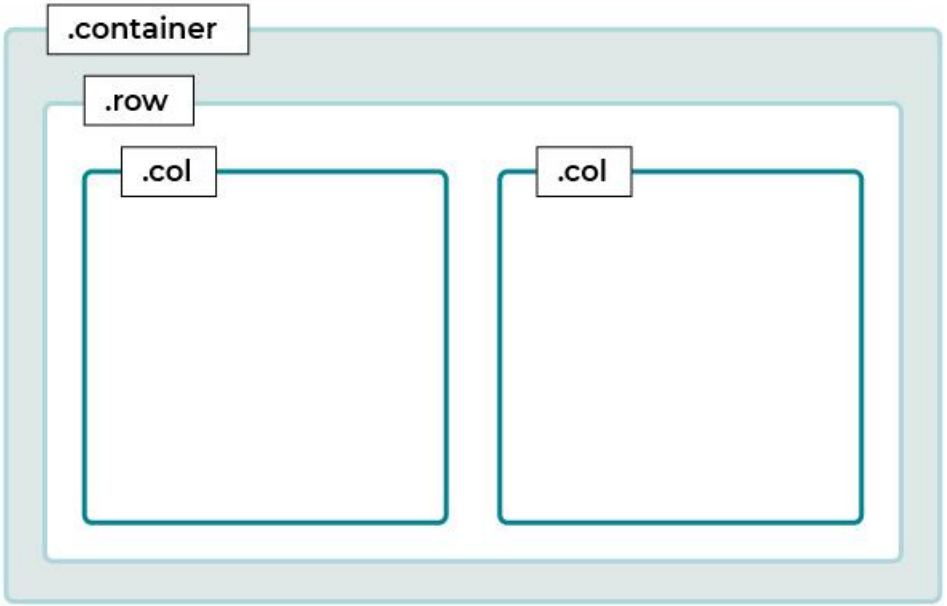
- Bootstrap is the most popular open-source CSS framework
- Provides ready-to-use UI components: buttons, navbars, cards…
- Predefined design language ensures consistent user interface
- Fastest way to build admin dashboards and internal tools
- Official website: https://getbootstrap.com/

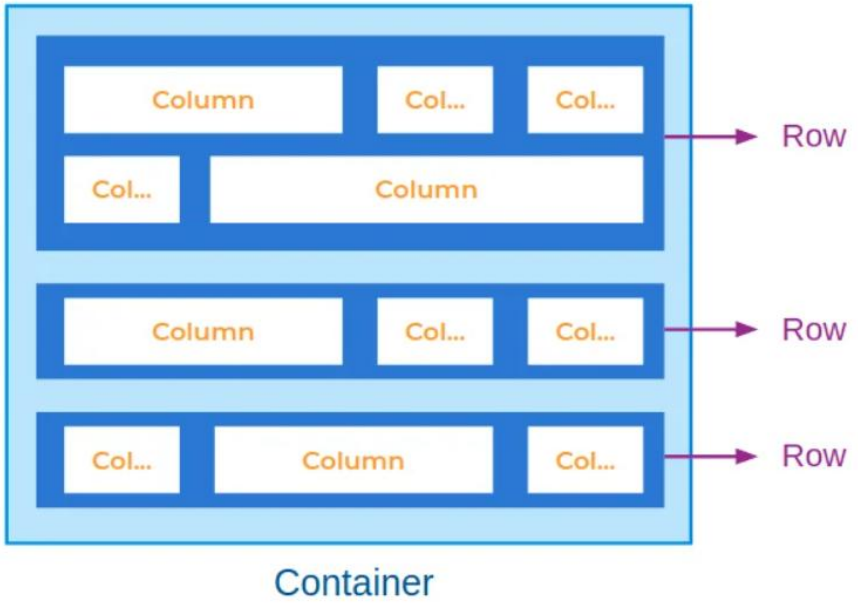
Example


getbootstrap.com/docs/5.3/components/buttons/
Docs
Examples
Icons
Blog
Search
CTRL K
v5.3
Components
Accordion
Alerts
Badge
Breadcrumb
Buttons
Button group
Card
Carousel
Close button
Collapse
Dropdowns
List group
Modal
Navbar
Navs & tabs
Offcanvas
Pagination
Placeholders
Popovers
Progress
Scrollspy
HTML
<button type="button" class="btn">Base class</button>
The .btn class is intended to be used in conjunction with our button variants, or to serve as a basis for your own custom styles.
If you are using the .btn class on its own, remember to at least define some explicit :focus and/or :focus-visible styles.
Variants
Bootstrap includes several button variants, each serving its own semantic purpose, with a few extras thrown in for more control.
Primary
Secondary
Success
Danger
Warning
Info
Light
Dark
Link
HTML
<button type="button" class="btn btn-primary">Primary</button>
<button type="button" class="btn btn-secondary">Secondary</button>
On this page
Base class
Variants
Disable text wrapping
Button tags
Outline buttons
Sizes
Disabled state
Link functionality caveat
Block buttons
Button plugin
Toggle states
Methods
CSS
Variables
Sass variables
Sass mixins
Sass loops

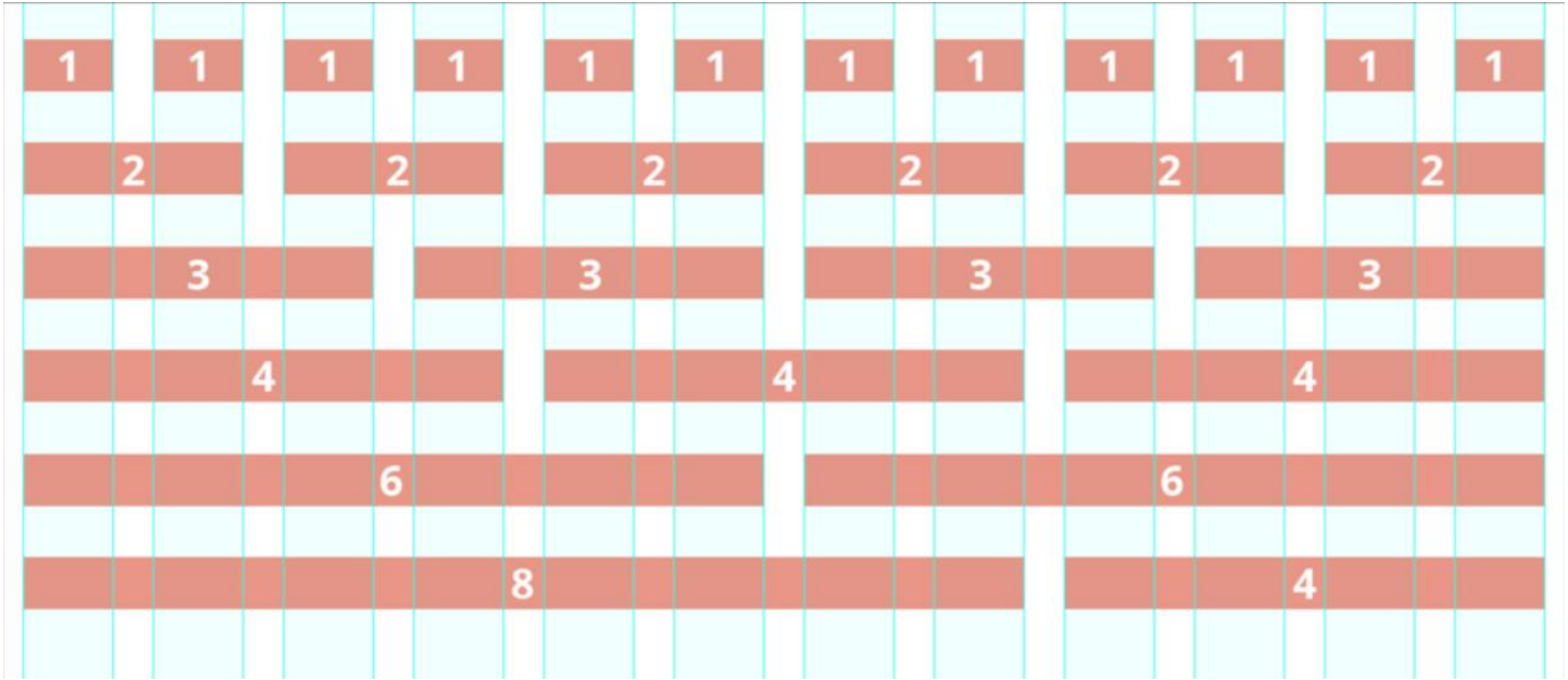
12-column responsive grid #1

- Layout is structured using containers, rows and columns
- .container: centers content with responsive horizontal padding
- .row: creates a horizontal group of columns with negative margins
- .col-*: specifies column span out of 12 available columns

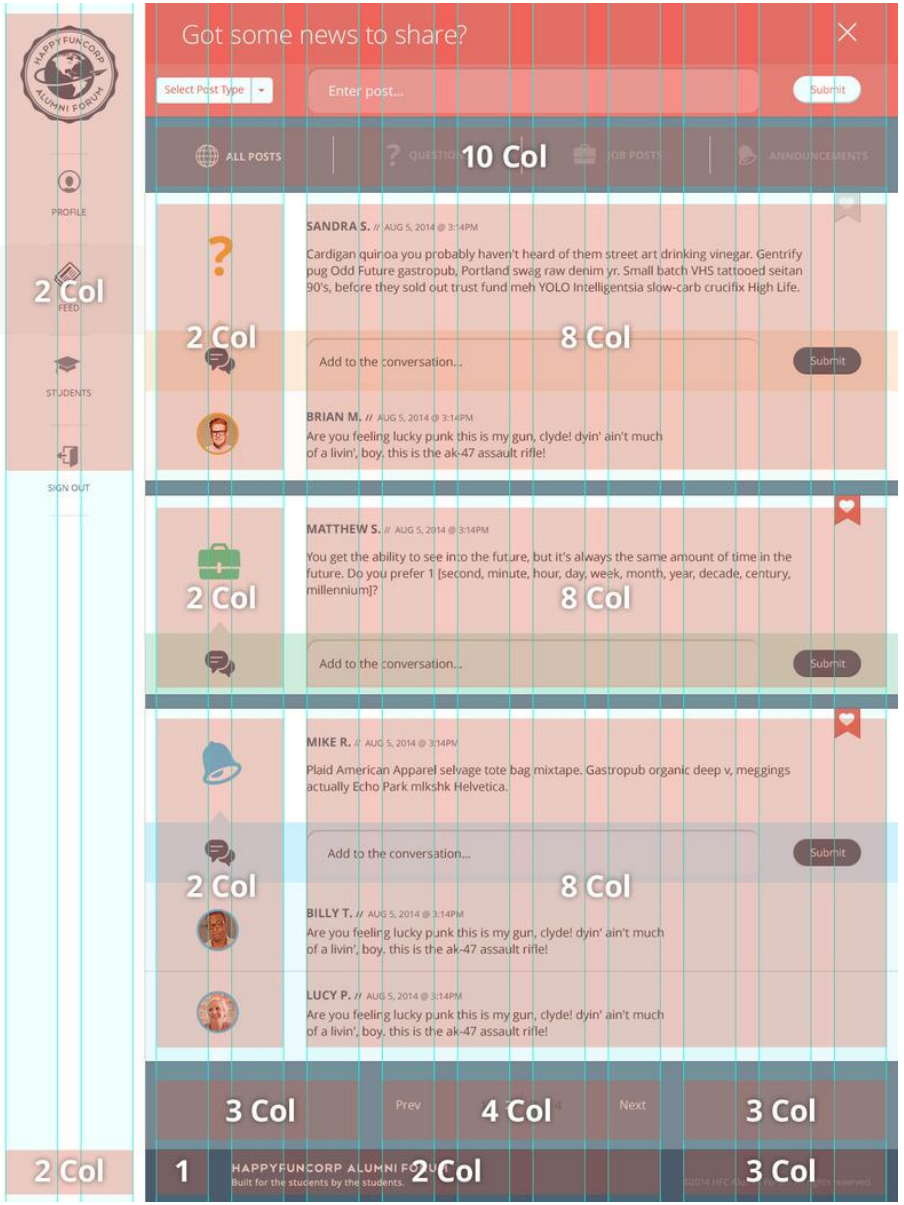




12-column responsive grid #2



12-column responsive grid #3



Breakpoints and layout rules #1

- Six default breakpoints: xs (<576px), sm, md, lg, xl, xxl
- col-12 col-md-6: full width on mobile, half width on tablet
- Offset and reorder columns with .offset-* and .order-*
- Built-in flexbox utilities align and distribute column space

Breakpoints and layout rules #2

xs	Extra small <576px	 portrait mobile
sm	Small ≥576px	 landscape mobile
md	Medium ≥768px	 portrait tablets *navbar collapse*
lg	Large ≥992px	 landscape tablets
xl	Extra large ≥1200px	 laptops, desktops, TVs

Bootstrap components & forms

Pre-styled UI controls and utility classes

Core UI components

- Buttons: .btn .btn-primary, .btn-secondary, .btn-outline-*
- Cards: .card, .card-header, .card-body for grouping content
- Navigation: .navbar, .nav-tabs, and responsive hamburger toggles
- Alerts and Badges: .alert .alert-success, .badge .bg-info

Form styling and validation

- .form-control styles inputs, textareas and date pickers
- .form-select provides styled native dropdown selectors
- .form-check structures checkboxes and radio buttons
- .is-invalid and .invalid-feedback display validation errors

Bootstrap utilities and JS widgets

- Spacing utilities: m-3, p-2, mt-auto, mx-auto
- Flex utilities: d-flex, justify-content-between, align-items-center
- Text and color: text-center, fw-bold, text-muted, bg-light
- Interactive widgets: Modal, Dropdown, Accordion require bootstrap.js

Integrating Bootstrap with Thymeleaf

- Quick setup: include Bootstrap CSS and JS bundles via CDN
- Self-contained setup: add webjars or static files in /static/css/
- Thymeleaf links: th:href='@{/css/bootstrap.min.css}'
- Clean integration with Spring MVC forms and table listings

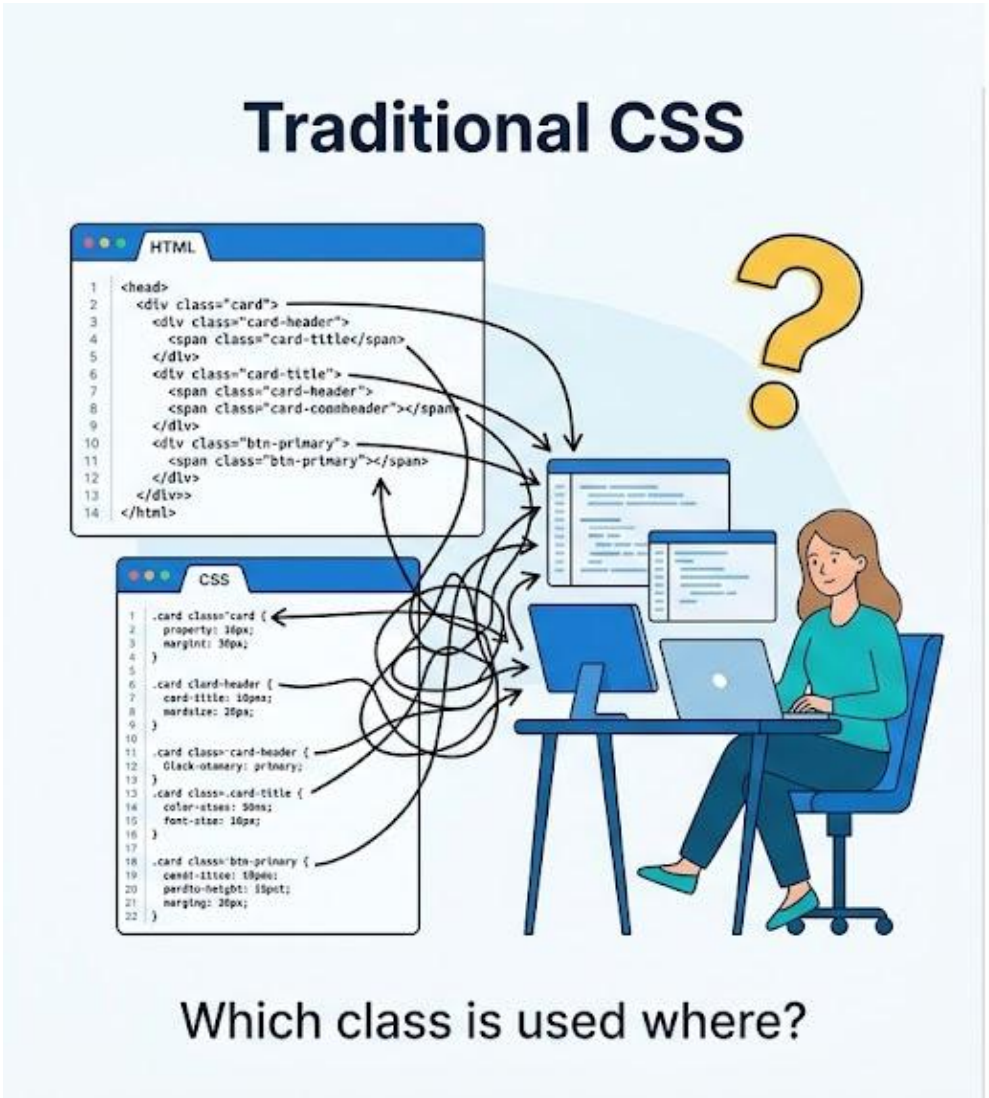
TailwindCSS: Utility-First Styling

Building bespoke designs without custom CSS

The utility-first philosophy

- Compose designs directly in HTML using low-level utility classes
- No opinionated pre-built components to override
- Never spend time inventing arbitrary CSS class names
- No separate CSS file growing indefinitely over time
- Official website: https://tailwindcss.com/

Traditional CSS vs TailwindCSS





Why choose TailwindCSS?

- Total design freedom: no two TailwindCSS websites look alike
- Tiny production bundle: purge removes all unused CSS rules
- Design tokens: strict spacing, color and typography scales
- Local changes: modifying a component cannot break other pages

Setup: CDN vs Tailwind CLI

- Play CDN (<script src='...'>): instant sandbox prototyping
- Tailwind Standalone CLI: official binary compiling CSS without Node.js
- Input CSS defines @tailwind base, components, utilities
- Watch mode scans Thymeleaf templates and regenerates output CSS

TailwindCSS syntax & responsiveness

Spacing, typography, flexbox and states

Spacing, sizing and colors

- Spacing scale: p-4 (padding 1rem), m-2, space-y-4
- Sizing: w-full, w-1/2, h-12, max-w-xl
- Colors: bg-blue-600, text-gray-800, border-gray-300
- Typography: text-lg, font-semibold, tracking-wide

Flexbox and Grid layout

- Flexbox: flex, flex-col, items-center, justify-between
- Flex gap: gap-4, gap-x-6 (clean spacing between items)
- CSS Grid: grid, grid-cols-1, md:grid-cols-3, gap-6
- No complex row/col nesting required

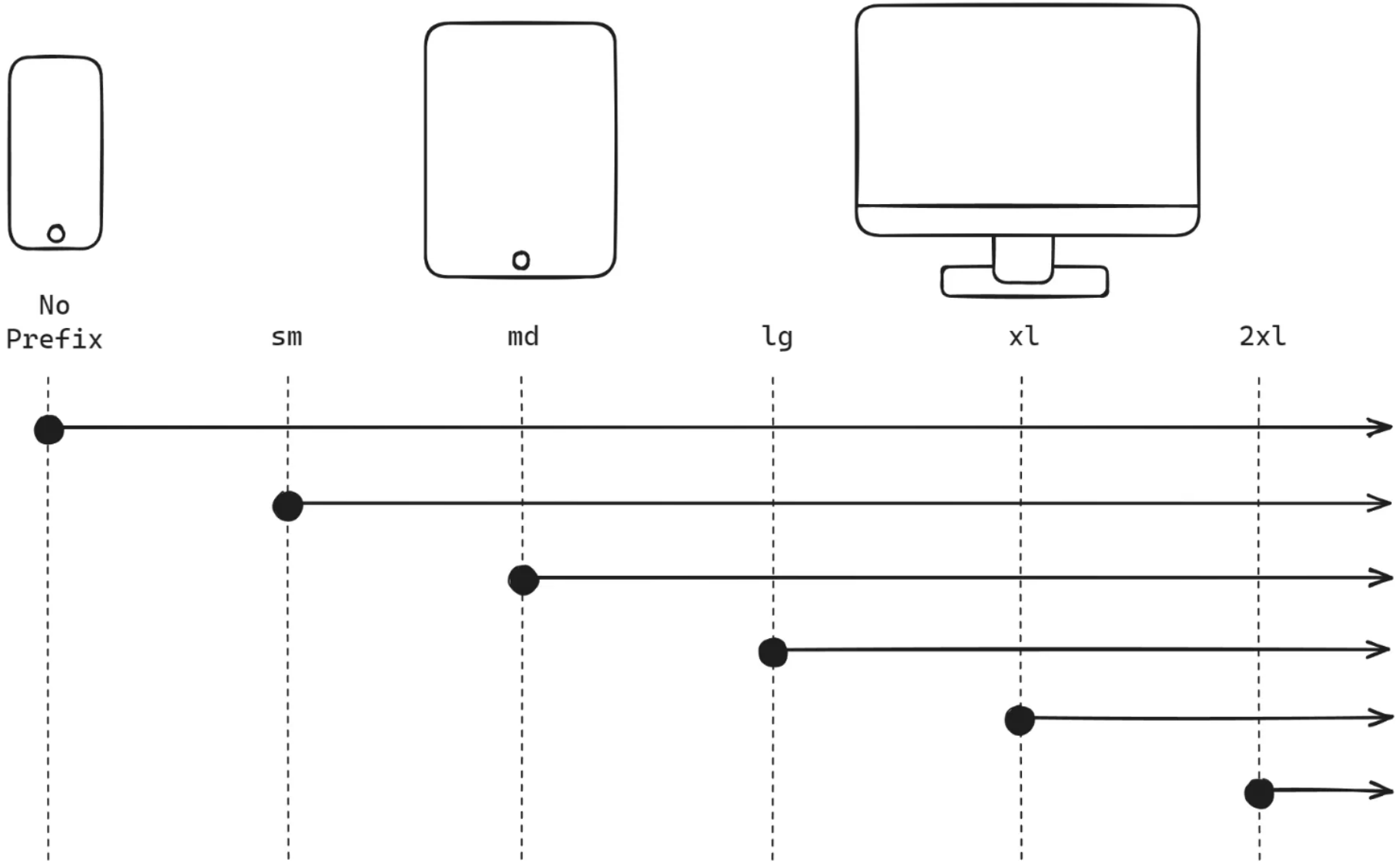
Mobile-first responsive design

- Default utility applies to all screen sizes (mobile-first)
- sm: 640px, md: 768px, lg: 1024px, xl: 1280px
- w-full md:w-1/2: full width on mobile, half width on desktop
- Responsive prefix applies conditionally when viewport matches

Hover, focus and state variants

- hover:bg-blue-700: interactive hover state change
- focus:ring-2 focus:ring-blue-500: accessible focus outline
- disabled:opacity-50: visual state for disabled controls
- dark:bg-gray-900: built-in dark mode support

Tailwind responsive design



Takeaway

- Bootstrap accelerates development through pre-built components.

- TailwindCSS offers limitless design freedom with utility classes and minimal production CSS.

Q&A – Thank you